

Artificial Neural Networks – Unit 7 Reflection

What

In this unit, I learned the fundamental concepts behind artificial neural networks, starting with the **single perceptron**, the simplest form of a neural network that makes decisions based on weighted inputs and an activation function.

I then explored the **single-layer perceptron**, using the **AND Boolean function** as an example. The AND function can be solved with a single layer because it is **linearly separable**, meaning a single decision boundary can correctly classify all possible inputs.

The unit also introduced the **multi-layer perceptron (MLP)** through the **XOR Boolean function**. Unlike AND, XOR is **not linearly separable**, so it cannot be solved by a single-layer perceptron. Adding hidden layers allows the network to learn more complex, non-linear relationships between inputs and outputs.

Another important topic was the role of **activation functions**, comparing the traditional **step function** with the **sigmoid function**. While the step function produces binary outputs, the sigmoid function provides smooth and differentiable outputs, making gradient-based learning possible.

Finally, I studied the key mechanisms that enable neural networks to learn, including **weights**, **loss functions**, **learning rates**, and **weight updates** through gradient descent.

So What

Implementing three types of perceptrons, as well as a perceptron for the **Boolean majority function** (attached in the e-portfolio), using only **NumPy**, helped transform abstract neural network concepts into practical and understandable models. Building the networks from scratch provided a deeper understanding of how layers, weights, activation functions, and learning algorithms interact to produce intelligent behavior.

More importantly, the implementation highlighted the iterative nature of neural network training through **gradient descent**:

1. **Forward Propagation**
Inputs are passed through the network to generate predictions.
2. **Compute Loss**
The prediction is compared with the actual target value to measure error.
3. **Backpropagation**
Gradients are calculated to determine how each weight contributed to the error.

4. **Update Weights**

Weights are adjusted using the gradients and learning rate to reduce the loss.

5. **Repeat**

The process continues over many iterations until the model converges to an acceptable solution.

Seeing these steps executed in code made it clear that learning in neural networks is not a single calculation but a continuous cycle of prediction, evaluation, and improvement. The exercise also demonstrated why multi-layer architectures are significantly more powerful than single-layer models when dealing with non-linear problems.

Now What

Implementing the perceptron for the **Boolean majority function** helped me appreciate how powerful even a simple neural network can be. In my implementation, the perceptron learned the target function in only **six training epochs**, showing how efficiently linearly separable problems can be represented and learned.

This exercise also highlighted the importance of choosing an appropriate model for a given problem. While the majority function can be represented compactly by a single perceptron using a set of weights and a threshold, a decision tree requires significantly greater complexity. In fact, theoretical results show that the majority function requires a decision tree of exponential size in the worst case (Russell, Norvig and Davis, 2010).

In future learning, I would like to explore different **activation functions**, such as ReLU and softmax, and compare how they affect model performance. I also want to investigate different **loss functions** and optimization methods beyond basic gradient descent, such as stochastic gradient descent and Adam optimization.

This unit has shown me that artificial neural networks are built from simple mathematical ideas, but when these ideas are combined in layers, they can solve complex problems. The knowledge gained from this unit gives me a stronger foundation for studying advanced machine learning and artificial intelligence topics in the future.